

Review

Mathematical modelling application in maritime Vessel: An analysis of bibliometric research publications from 1962 to 2023 utilizing Scopus databases

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ABSTRACT

Mathematical modelling is a crucial tool in maritime research, optimizing vessel performance, safety, and environmental sustainability. This study provides a 61-year bibliometric analysis (1962–2023) of mathematical modelling in maritime vessels using Scopus data. It highlights global research trends, key contributors, and emerging themes such as Digital Twin technologies and artificial intelligence. While identifying research gaps, the analysis reveals the field's response to challenges like decarbonization and automation. Insights are provided for researchers, policymakers, and industry stakeholders to enhance sustainable practices and innovation. The findings emphasize the evolving role of mathematical modelling in addressing critical maritime issues, offering a roadmap for future advancements.

1. Introduction

The evolution of mathematical modelling in maritime vessels (MMV) since the 1960s has been significantly influenced by advancements in computational methods and the increasing complexity of vessel designs. As marine vessels play a crucial role in global commerce, optimizing their performance and reducing their environmental impact have become major priorities. Mathematical modelling serves as a vital tool for simulating various aspects of vessel performance, including hydrodynamic behaviour, structural integrity, and operational efficiency. Over the past six decades, foundational studies have progressively shaped the development of MMV research, leading to innovations in hydrodynamic simulation, artificial intelligence (AI) integration, and digital twin technologies. These advancements have facilitated the transition from early physics-based deterministic models to data-driven, machine learning-enhanced methodologies, enabling greater predictive accuracy and real-time operational insights.

As regulatory pressures on emissions reduction and fuel efficiency have intensified, the role of mathematical modelling has expanded beyond traditional ship performance evaluations to include environmental impact assessments and fuel optimization techniques (Corbett and Winebrake, 2008; Winebrake et al., 2009). In particular, mathematical models now support the integration of environmentally benign

fuels, such as liquefied natural gas (LNG) and alternative gaseous fuels, to meet stringent environmental policies (Livaniou and Papadopoulos, 2022; Livanos et al., 2014; Wan et al., 2015). These models allow researchers and engineers to analyse how vessels interact with their marine environment, leading to improvements in safety, fuel economy, and emissions reduction (Ghaemi and Zeraatgar, 2022; Ruiz Zardoya et al., 2023; Sunarsih et al., 2015).

1.1. Foundational studies in maritime mathematical modeling

1.1.1. Ship maneuvering and hydrodynamic modeling

One of the earliest standardized approaches to ship maneuvering predictions was the Maneuvering Mathematical Modeling Group (MMG) Standard Method, introduced by Yasukawa and Yoshimura (2015), provided a foundational methodology for ship maneuvering predictions, enabling systematic hydrodynamic analysis. This approach has been widely adopted and refined to address challenges such as vessel stability and efficiency in diverse maritime environments. Similarly, Computational Fluid Dynamics (CFD) has emerged as a transformative tool in maritime engineering, revolutionizing the simulation of complex flow dynamics around ship hulls (Sutulo and Guedes Soares, 2014). CFD techniques now serve as a core analytical method for evaluating hull designs, propeller efficiency, and wave resistance, significantly

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improving performance optimization.

Further advancements in hydrodynamic modelling have allowed researchers to analyse vessel interactions with waves, currents, and extreme weather conditions, leading to improvements in ship safety, manoeuvrability, and fuel efficiency (Araki et al., 2012). As the field evolved, machine learning-enhanced CFD simulations emerged, allowing for real-time hydrodynamic assessments in autonomous shipping applications.

1.1.2. Energy efficiency and emissions reduction

The growing focus on fuel efficiency and emissions reduction has positioned mathematical modelling as a key tool for optimizing ship energy consumption and environmental performance. Shi et al. (2020) introduced a machine learning-based emissions monitoring model, which significantly improved fuel efficiency assessments while ensuring compliance with IMO regulations. This study demonstrated that integrating real-time operational data with predictive algorithms could reduce fuel consumption and greenhouse gas emissions.

The adoption of alternative propulsion systems has also been a major focus in MMV research. Ruiz Zardoya et al. (2023) developed optimization frameworks for LNG-powered vessels, providing critical insights into fuel efficiency, cost, and emissions trade-offs. These models support the transition towards hydrogen and ammonia-based propulsion technologies, aligning with global efforts to decarbonize maritime transportation. Moreover, simulation tools have played a critical role in evaluating the impact of various fuel types on engine performance and emissions, helping to identify optimal operational strategies (Liang et al., 2021; Ye et al., 2021).

1.1.3. AI and machine learning in ship modelling

The integration of artificial intelligence (AI) and machine learning has significantly expanded the capabilities of maritime vessel modelling, enhancing predictive analytics and real-time decision-making (Nguyen et al., 2023). pioneered the use of artificial neural networks (ANNs) for ship fuel consumption prediction, demonstrating higher accuracy than traditional statistical models. Their work exemplifies the transition from deterministic, physics-based models to adaptive AI-driven methodologies, improving voyage optimization and predictive maintenance.

Additionally, the application of AI in port operations and scheduling models has shown promise in reducing overall fuel consumption and emissions (Dadashi et al., 2017; Yuan et al., 2021). The continuous expansion of machine learning applications in MMV has led to breakthroughs in automated navigation, predictive maintenance, and fuel efficiency forecasting, making AI-driven modelling an essential tool for sustainable maritime logistics.

Furthermore, Digital Twin technology, which enables real-time vessel monitoring, has become a dominant research theme in maritime performance modelling. Stoumpos et al. (2020) introduced a dual-fuel engine Digital Twin model, demonstrating its effectiveness in predicting maintenance needs and optimizing engine performance. The continued development of Digital Twin systems is expected to support automation and remote diagnostics in next-generation shipping operations.

1.2. Evolution of mathematical modelling techniques in maritime research

The progression of mathematical modeling in maritime vessel research can be categorized into three major phases, reflecting advancements in computational power, data availability, and industry demands:

- i) Pre-2000: Physics-Based and Deterministic Models
 - Hydrodynamic equations and empirical models were developed to assess ship stability, resistance, and manoeuvrability.
 - The MMG model provided a standardized framework for manoeuvring simulations.

- ii) 2000–2015: Computational Fluid Dynamics and Optimization Techniques
 - CFD simulations enhanced the accuracy of hull resistance and propulsion system modelling.
 - Optimization techniques were introduced to reduce fuel consumption and emissions.
- iii) Post-2015: AI, Machine Learning, and Digital Twin Technologies
 - Neural networks and reinforcement learning algorithms improved predictive modelling for ship energy efficiency.
 - Digital Twin models enabled real-time performance monitoring, predictive maintenance, and autonomous ship control.

As vessel configurations become increasingly complex, there is a growing need for more advanced and precise modelling methodologies. Current research focuses on addressing climate change challenges and fossil fuel dependency, ultimately contributing to the development of safer, more efficient, and environmentally sustainable marine vessels (Breyer et al., 2019; Satpathi et al., 2017). The future of MMV will likely emphasize integrating renewable energy sources and next-generation propulsion systems, further highlighting the critical role of advanced mathematical modelling in shaping sustainable maritime innovation (Khalili et al., 2019; Kumar et al., 2019).

Although substantial progress has been made in MMV research, a comprehensive bibliometric analysis of the field over the past 60 years remains to be conducted. This paper aims to address this gap by exploring the evolution of MMV research, identifying key trends, influential contributors, and emerging themes. Bibliometric analysis is a valuable method for understanding the progression of scientific knowledge within a field, allowing for the identification of research patterns, leading authors, institutions, and the dominant research topics that have shaped MMV development over time.

This study also aims to provide insights into how the field of MMV has responded to emerging global challenges such as climate change, resource scarcity, and the need for greener technologies. By analysing trends in MMV research, this paper will highlight how the community has contributed to global efforts aimed at improving vessel efficiency, reducing emissions, and integrating sustainable practices in maritime operations. Furthermore, the analysis will offer valuable guidance for future research and collaboration opportunities in the field, with implications for both academic researchers and industry stakeholders.

The objectives of this paper are:

- i) To analyse the distribution patterns of MMV journal articles over time;
- ii) To showcase contributions of prominent authors, leading countries, and the most productive academic institutions;
- iii) To emphasize common terminology and research themes within MMV;
- iv) To assess the influence of countries based on key applications of MMV; and
- v) To provide insights into possible collaboration opportunities and future research directions.

Ultimately, this study aims to provide a roadmap for advancing MMV research in line with global priorities for sustainability and innovation in maritime technology.

2. Methodology

Grounded in a bibliometric analysis, this study investigates the evolution of scholarly contributions pertaining to marine vessel mathematical modelling from the earliest recorded date in the Scopus database in 1962 up to 2023, with the objective of evaluating global advancements and scholarly interests. The extended timeline allows for an understanding of the foundational studies that shaped mathematical modelling in maritime vessels. However, the study primarily emphasizes

advancements post-2000, a period characterized by significant technological innovations such as machine learning, Digital Twin technologies, and energy optimization models. Older trends are contextualized to illustrate the progression of methodologies and their influence on modern practices. A bibliometric analysis provides a distinctive perspective on the significance of research trajectories, unlike conventional review articles that typically focus on outlining recent advancements, progress, and future recommendations within a specific research area. For this study, Scopus was selected as the sole data source due to its extensive coverage of high-quality, peer-reviewed journals and its reliable citation metrics, which provide a solid foundation for bibliometric analysis (Pranckutė, 2021; Sweileh, 2019). The decision to focus exclusively on Scopus was made to maintain a clear and focused scope within the constraints of this research. However, it is acknowledged that Scopus, while comprehensive, may not capture all relevant contributions, particularly those found in other databases such as Web of Science and IEEE Xplore, which include unique content like conference proceedings and specialized journals.

2.1. Data source and search methodology

On September 19, 2024, Data collection was conducted using the Scopus database, concentrating on the usage of MMV in both the titles and abstracts of the articles. The search string employed was "TITLE-ABS ('mathematic* model*' AND ship OR vessel)". After modifying the search parameters to limit the results exclusively to journal articles published between 1962 and 2023, the resultant dataset was condensed to 3425 documents. In summary, these outcomes were filtered based on criteria including year, source, authorship, subject area, institutional affiliation, geographic location and document classification. It is important to note that the latest search parameters have systematically excluded standard review articles that included terms such as potentiometric, progress, updates, overview, review, and bibliometric. The search string also included the Electronic Identifier (EID) for review articles from Scopus to ensure that only relevant research articles were retained. The accuracy of each abstract of the review articles was then subjected to manual verification. Subsequently, the extraction process aimed at obtaining the information pertaining to single-country publications (SCP) for a designated nation was executed by constraining the search outcomes via the AFFILCOUNTRY code. The evaluation of rankings to develop a structured tabulation of data was conducted utilizing bibliometric indicators accessible on the Scopus platform, including total citations, total publications, h-index Field-Weighted Citation Impact (FWCI), and Cite Score. The aggregate data collection, filtering, and elimination procedure is illustrated in Fig. 1. Further

information on the primary search strings is provided in Table 1.

This query ensures analysis focuses exclusively on journal articles, excluding conference papers and review articles, to ensure high-quality, peer-reviewed sources. The scope covers engineering, mathematics, and materials science disciplines, with a rigorous refinement process. Duplicate entries, such as preprint and final versions, were merged, while studies labeled as "review," "overview," "progress," or "bibliometric" were excluded. Additionally, manual abstract screening was performed to ensure relevance to mathematical modeling and maritime vessels.

2.2. Bibliometric maps

In the present investigation, a bibliometric map was generated utilizing VOSviewer software (version 1.6.18, Centre for Science and Technology Studies, Leiden University, Netherlands). This software was selected as the principal instrument for the analysis owing to its no-cost usage policy and its extensive compatibility with databases including Scopus, Web of Science, Dimensions, Microsoft Academic, Lens, and PubMed. The bibliometric map was constructed from 3425 scholarly articles extracted from the Scopus database. Within this map, nations and author keywords are depicted as entities, wherein the strength of the connection between two elements is computed by the software and represented as a readily comprehensible numerical value. A higher numerical value signifies a more robust association between the two elements. A minimum keyword occurrence threshold of five was set to capture significant trends without including isolated mentions. This threshold was validated through sensitivity testing at values of 3, 5, and 10, confirming stability at 5. Furthermore, the top 50 most-cited studies were identified to analyse their influence on newer publications, ensuring that high citation counts correlated with significant technical advancements.

Two distinct categories of bibliometric maps were constructed utilizing the dataset derived from the 3425 research articles. In the context of the co-authorship map, the intensity of the connections was ascertained through the quantity of articles that involved authors associated with diverse nations, and the cumulative link strength illustrated the comprehensive co-authorship potency of a nation. Nonetheless, VOSviewer exhibits specific constraints when performing an in-depth examination of the interrelations among countries. Conversely, the co-occurrence map evaluated the incidence of articles wherein two keywords were concurrently present. Both maps utilized a comprehensive counting methodology for data visualization. Additional information regarding the functionalities of the software can be located in its user manual. Within the co-authorship map, publication affiliations were

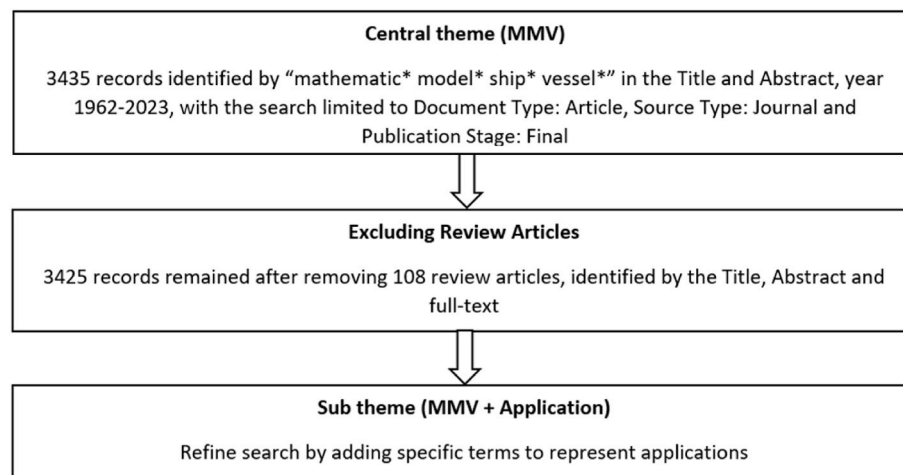


Fig. 1. Flowchart illustrating the methodology for the collection of publication data.

Table 1
The search strings employed in this study.

No.	Refinement Process	Search String	No. of Document
1	Central theme	TITLE-ABS (mathematic* AND model*) AND ship OR vessel	28037
2	Central theme – Exclude year 2024. Limit to journal article.	TITLE-ABS (mathematic* AND model* AND ship OR vessel) AND (LIMIT-TO (SUBJAREA, "ENGI") OR LIMIT-TO (SUBJAREA, "MATH") OR LIMIT-TO (SUBJAREA, "MATE")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English")) AND (EXCLUDE (PUBYEAR, 2024)) AND (LIMIT-TO (PUBSTAGE, "final"))	3435
3	Possible review article	TITLE-ABS (mathematic* AND model* AND ship OR vessel) AND (LIMIT-TO (SUBJAREA, "ENGI") OR LIMIT-TO (SUBJAREA, "MATH") OR LIMIT-TO (SUBJAREA, "MATE")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English")) AND (EXCLUDE (PUBYEAR, 2024)) AND (LIMIT-TO (PUBSTAGE, "final")) AND (TITLE (recent OR progress OR review OR critical OR revisit OR advance* OR highlight OR perspective OR prospect OR trends OR bibliometric OR scientometric OR insights OR overview OR "state of the art" OR challenges OR updates) OR ABS (progress OR review OR bibliometric OR scientometric))	108 Suspect
	Research article only	TITLE-ABS (mathematic* AND model* AND ship OR vessel) AND NOT EID (2-s2.0-0020785905 OR 2-s2.0-85063997834 OR 2-s2.0-84947034026 OR 2-s2.0-0027789767 OR 2-s2.0-85040279783 OR 2-s2.0-85101469109 OR 2-s2.0-85023751151 OR 2-s2.0-85018439332 OR 2-s2.0-85124981723 OR 2-s2.0-85104414090) AND (LIMIT-TO (PUBSTAGE, "final")) AND (LIMIT-TO (SUBJAREA, "ENGI") OR LIMIT-TO (SUBJAREA, "MATH") OR LIMIT-TO (SUBJAREA, "MATE")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (EXCLUDE (PUBYEAR, 2024)) AND (LIMIT-TO (LANGUAGE, "English"))	3425

employed to discern collaborative publications, whereas the co-occurrence map concentrated on collective publications identified by the simultaneous presence of keywords pertinent to MMV.

2.2.1. Co-authorship analysis

For the objective of investigating co-authorship within this research, the keywords pertinent to the authors' countries from a total of 3425 scholarly articles were meticulously quantified and represented visually. The VOSviewer software, in its initial mapping phase, identified 99 distinct countries and established connections to 2007 individual authors. Following a filtration process, the conclusive tally was reduced to 85 countries, as seven nations were omitted due to the fact that the keywords extracted from the publications corresponded to institutions rather than specific national affiliations. Subsequently, countries were

systematically categorized into various clusters according to their geographical regions, including Europe, Asia, Oceania, Americas and Africa. The maps were constructed utilizing the full counting methodology and were presented in a network visualization format, thereby facilitating a lucid depiction of co-authorship dynamics and collaborative networks across global territories within the MMV research domain.

2.2.2. Co-occurrence analysis

Utilizing the 2007 author keywords extracted from the 3425 research articles generated from the primary search string within the Scopus database, a bibliometric representation of co-occurring author keywords was constructed. The selected keywords underwent a refinement process to eradicate duplicates and to guarantee that each keyword possessed a unique semantic interpretation. Redundant or analogous keywords were standardized through the application of a thesaurus file. For instance, terms such as "maneuvering," "manoeuvring," and "maneuvering" were consolidated into a singular classification of "manoeuvring." A revised iteration of the map was developed incorporating 1992 author keywords. Nevertheless, only 101 keywords persisted when the threshold for the minimum frequency of author keyword occurrences was established at 5. The resultant map was exhibited in overlay visualization mode, wherein the coloration of the items and the strength of the links signified the average publication year of the articles that encompassed the specified keywords.

2.3. Contributor and effects of MMV

The study compares the main theme (keyword co-occurrence) with sub-themes (total number of publications) to investigate the relationship between different contributor types and the effects of MMV. As an illustration, "vessel performance optimization" is cited as a major factor in the development of maritime vessel design. The optimization of vessel performance has become a critical area of research in maritime engineering, particularly in the context of energy efficiency and emissions reduction. One of the primary strategies for achieving these goals involves the integration of advanced data analytics and machine learning techniques into ship performance monitoring systems.

For instance, H. Li et al. (2023) develops a data-driven machine learning framework for ship CO₂ emission estimation, utilizing AIS trajectory segmentation to enhance prediction accuracy. Such advancements align with studies like Bui & Perera (2019) which emphasize the need for advanced data analytics to tackle big data challenges in maritime operations. The integration of machine learning for predictive modelling, as seen in Handayani et al. (2023), further supports this goal by demonstrating high-accuracy fuel consumption prediction models, directly contributing to emissions reduction efforts.

Furthering this integration, research by Song et al. (2024) presents a mixed-integer linear programming model aimed at optimizing vessel speed and berth allocation to improve maritime efficiency. Similarly, Jia et al. (2023) proposes an Adaptive Ant Colony Optimization (AACO) algorithm for vessel traffic scheduling in restricted harbour basins, enhancing efficiency by reducing total scheduling and waiting times. These studies collectively highlight the role of optimization algorithms in balancing vessel speed, port constraints, and fuel efficiency to minimize environmental impact.

Additionally, techniques for trajectory prediction using advanced algorithms, like Bi-LSTM (Yang et al., 2022), have been instrumental in improving traffic management and collision avoidance. Expanding on this, Duan et al. (2022) introduces a semi-supervised deep learning approach (SSL-VTC) for vessel trajectory classification, utilizing convolutional neural networks (CNNs) to extract feature representations and outperform traditional supervised learning methods. These approaches contribute to safer and more efficient maritime operations through enhanced navigation prediction and decision-making.

Moreover, the integration of robotics in port logistics and vessel maintenance, as explored by (Sahoo et al., 2024), suggests that

automated systems can streamline operations and improve efficiency. The importance of electrification and alternative fuels in ship design has also been underscored, emphasizing their role in reducing carbon emissions (Rauca and Batrinca, 2023). Building on this, Gao et al. (2023) introduces a ship collision avoidance model that incorporates carbon emission constraints, integrating a modified potential field ant colony algorithm for real-time, dynamic path optimization. By leveraging AIS data, this model enhances navigation safety while ensuring environmental sustainability.

Furthermore, the development of marine engine-centered data analytics frameworks, as discussed by (Perera and Mo, 2017) aligns with these strategies by enforcing emission control measures and improving vessel navigation performance. In parallel, Liu et al. (2021) employs machine learning techniques to analyse maritime accidents, identifying risk patterns and offering data-driven recommendations to enhance vessel safety. This focus on risk prevention aligns with broader efforts in predictive analytics and maritime safety management, ensuring proactive hazard mitigation in coastal waters.

As a result, research publications including the term “vessel performance optimization” were examined. We analyzed the number of research articles from the top five nations contributing to the advancement and application of MMV, highlighting key studies that enhance operational efficiency, emissions reduction, and maritime safety.

3. Results and discussion

3.1. Production of research article and research interest

3425 research articles on MMV have been published in the Scopus database over a 61-year period, from 1962 to 2023. Fig. 2 shows the expansion of MMV papers in the Scopus database. The fascination with mathematical modelling for marine vessels among researchers is evident in the total number of publications, which has shown a steady increase each year since the beginning of 1962. There is no decline in publications at all throughout the years. The overall trend continued to rise. The accessibility of research articles also plays a role in promoting interest, as 1960 out of 3425 articles, or 57.25 %, are open-access papers. Furthermore, the predominance of articles published in English (3,377, or 98.6 %) facilitates broader engagement within the research community.

Mathematical modelling in marine vessels is a topic that has been studied in many academic fields and is still important in the current scientific and technical environment. A review of every paper on this topic in the Scopus database shows that most of the publications, 1205

research articles (35.2 %), fall under the category of Engineering. Generally, advancements in computational technology have been closely linked to the development of mathematical models for marine engineering, as enhanced modelling techniques allow for more accurate simulations of vessel performance and efficiency. As highlighted by recent studies, ongoing innovations in computational methods continue to drive research in the field of MMV.

3.2. Preferred article

Based on information gathered from the top ten MMV journals (Table 2), Elsevier publishes four of these publications. The titles of these journals include “Ocean Engineering” “Journal of Theoretical Biology,” “Journal of Biomechanics,” and “Applied Ocean Research.” Springer Nature has two journals within the list. The total productivity (TP) value of the top-ranked journal, “Ocean Engineering,” is 213 (6.2 %), meaning that articles in this journal are directly related to mathematical modelling in marine vessels. Out of ten, there are 5 journal that related to marine optimization which are Journal of Marine Science and Engineering, Applied Ocean Research, Polish Maritime Research and International Shipbuilding Progress. “Applied Ocean Research” and “Ocean Engineering” have second and third highest Cite score rating with 8.7 and 7.3 respectively with the both most cite article are related to modelling of ship manoeuvrings reflecting an active readership and consistently provides valuable insights into the fields of marine engineering and vessel optimization. A higher Cite Score indicates greater journal value and is often a key criterion for authors to select the institution for their publication.

3.3. Prominent countries and institutions

This analysis reveals 10 countries and their most productive institutions that have significantly influenced the publication of MMV-related works as shown in Fig. 3. With a total publication count (TPC) of 669 and a single-country publication (SCP) rate of 73 %, China is identified as the prominent country for MMV output of research. Other notable contributors include United States, United Kingdom, India, Italy, Russian Federation, Japan, Poland, France and Germany with majority of country with TPC of over 100 articles. Among these, Russian Federation exhibits the highest SCP value at 80 %, indicating a lowest level of intra-country collaboration compared to others. A higher SCP value suggests that a significant number of authors from different nations are linked to a single country; however, this can also discourage researchers from broadening their networks and engaging in

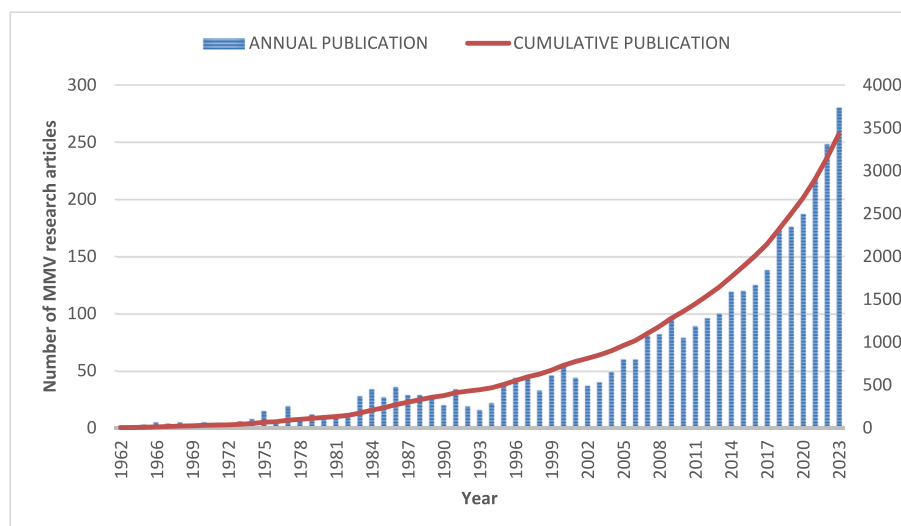


Fig. 2. The total number of publications per year and over time on related mathematical modelling of vessel indexed in Scopus from 1962 until 2023.

Table 2

The top ten journals with the highest number of citations for MMV works between 1962 and 2023.

RANK	JOURNAL	TP (%)	CITE SCORE 2023	THE MOST CITE ARTICLE	TIMES CITED	FWCI	PUBLISHER
1	Ocean Engineering	6.2 (213)	7.3	An algorithm for offline identification of ship manoeuvring mathematical models from free-running tests (Sutulo and Guedes Soares, 2014)	146	4.04	Elsevier
2	Journal Of Marine Science And Engineering	2.3 (79)	4.4	Dynamic load and strength determination of carrying structure of wagons transported by ferries (Lovska et al., 2020)	36	2.67	Multidisciplinary Digital Publishing Institute (MDPI)
3	Polish Maritime Research	1.5 (50)	3.7	Game control methods in avoidance of ships collisions (Lisowski, 2012)	27	0.77	Walter de Gruyter
4	Journal of Theoretical Biology	1.3 (46)	4.2	Mathematical modelling of dynamic adaptive tumour-induced angiogenesis: Clinical implications and therapeutic targeting strategies (McDougall et al., 2006)	326	9.51	Elsevier
5	Bulletin of Mathematical Biology	1.3 (44)	3.9	Continuous and discrete mathematical models of tumor-induced angiogenesis (Anderson, 1998)	898	4.99	Springer Nature
6	Journal of Biomechanics	1.1 (38)	5.1	Arterial viscoelasticity: A generalized model. Effect on input impedance and wave travel in the systematic tree (Westerhof and Noordergraaf, 1970).	132	–	Elsevier
7	IEEE Access	1.1 (37)	9.8	Zebra Optimization Algorithm: A New Bio-Inspired Optimization Algorithm for Solving Optimization Algorithm (Trojovska et al., 2022)	156	19.15	IEEE
8	Applied Ocean Research	1.1 (36)	8.7	Measures to diminish the parameter drift in the modeling of ship manoeuvring using system identification (Luo and Li, 2017)	57	3.15	Elsevier
9	International Shipbuilding Progress	1.0 (33)	2.6	Simulation and stability of ship towing (Bernitsas and Kekridis, 1985)	34	–	IOS Press
10	Annals of Biomedical Engineering	0.9 (32)	7.5	Cationic nanoparticles have superior transvascular flux into solid tumors: Insights from a mathematical model (Stylianopoulos et al., 2013)	118	3.74	Springer Nature

**Fig. 3.** The most productive country and universities according to MMV publications.

collaborative efforts. Apparently, Russia, China, India and Poland are four country in the list that have an SCP value exceeding 70 %.

The Dalian Maritime University in China has an impressive

publication record, with a total productivity index (TPI) of 84. This can be attributed to these institutions being recognized as global centres of scientific research, consistently producing high-quality articles.

Additionally, the second highest of (TPI) is drop down quite significant for other institution compare to Dalian Maritime University which is 23 and hold by two institutions namely Russian Academy of Sciences and Gdynia Maritime University.

The MMV discipline has shown a great interest in co-authorship as well as the volume of production from research institutes through publications. Fig. 4 displays the bibliometric map of nations based on co-authorships created with VOSviewer. This map was produced using data from 93 nations, illustrating the connections between them. The thickness of the connecting lines indicates the strength of the tie between two countries. Different colours indicate continents: blue for Europe, red for Asia, green for Middle Asia and yellow for America. Notably, 40 countries in Europe, 34 countries in Asia, 19 countries in America, and 4 countries in Africa are the origins of the MMV publications. As depicted in Fig. 4, the size of the circles surrounding each country's name varies according to the quantity of publications, with China having largest frame due to its contribution of 669 out of 3425 total publications follow by United States as the second with 534 total publications and United Kingdom with 285 total publications. Conversely, countries like Slovakia Bangladesh, Trinidad and Tobago and Botswana are represented with only one connection to another country.

The evolution of MMV research parallels the maritime industry's response to global challenges, including decarbonization and automation. Historical trends show a steady transition from single-objective optimization (e.g., vessel speed) to multi-objective frameworks encompassing fuel efficiency, emissions reduction, and safety. For instance, LNG and hybrid propulsion systems have gained prominence as viable alternatives to conventional marine fuels, aligning with IMO's strict emission standards (Livaniou and Papadopoulos, 2022). Regionally, China has emerged as a leader in research output, driven by robust investments in autonomous technologies. Conversely, Europe emphasizes sustainability initiatives, particularly in emission reduction and renewable energy integration (Chen and Yen, 2023). This regional disparity reflects varying institutional priorities, influenced by geopolitical and cultural contexts.

3.4. Prominent author

The most prominent authors in the MMV research field are listed in Table 3. According to the data, three out of ten authors are from the United Kingdom, two are from Belgium, and the remaining authors hail

from Portugal, India, United States, Taiwan and Italy. Most authors are recognized as first authors in research articles dating back to 1972. Additionally, they prioritize publishing research articles in marine engineering and mathematical modelling. Table 3 lists the most prominent authors in the field of MMV research. Based on the statistics, Portugal, India, the United States, Taiwan, and Italy are the remaining writers, while three out of ten are from the United Kingdom and two are from Belgium. Research articles from as far back as 1972 identify the majority of authors as first authors. They also place a high priority on publishing research papers in mathematical modelling and marine engineering.

Among the total of 151 MMV research articles, 20 were contributed by Chaplain, M.A.J. from University of St Andrews, United Kingdom with the highest citation of 1218. In second place is Byrne, H.M. who has published 18 research articles. As for the third place, Guedes Soares, C. with the total publication of 17, possesses the highest H-index of 99 with total citation of 532 although ranked third. The h-index value typically evaluates an author's output and the impact of their writings on a given study area. From 2019 to 2023 most contributed topic by Guedes Soares are Ship Collision; Waterway; Engineering; Wave Energy; Significant wave height; Climate change; Wind turbine; Offshore Oil Well Production and Hydrodynamics. Other prolific authors gain a total citation above 100 are include Vantorre, M. from Belgium, Misra, J.C. from India, Fang, M.C from Taiwan and Fasano, A. from Italy who have produced 114, 276, 282 and 104 respectively. Generally, the authors listed in Table 3 are affiliated with research institutes or laboratories specializing in marine engineering and mathematical modelling. As a result, a variety of research efforts, including the development of software and hardware, are regularly reflected in their publications as well as analysis of data related to MMV.

3.5. Author keywords

VOSviewer found 8600 author keywords associated with MMV research based on the collected database. Following the initial examination of the results, terms having congeneric or similar meanings were re-labeled to prevent recurrence. Setting the minimal number of occurrences for the bibliometric maps at nine (9) resulted in 83 terms meeting the criteria.

3.5.1. Terminology and concept

The term "mathematical model" was the most common in MMV

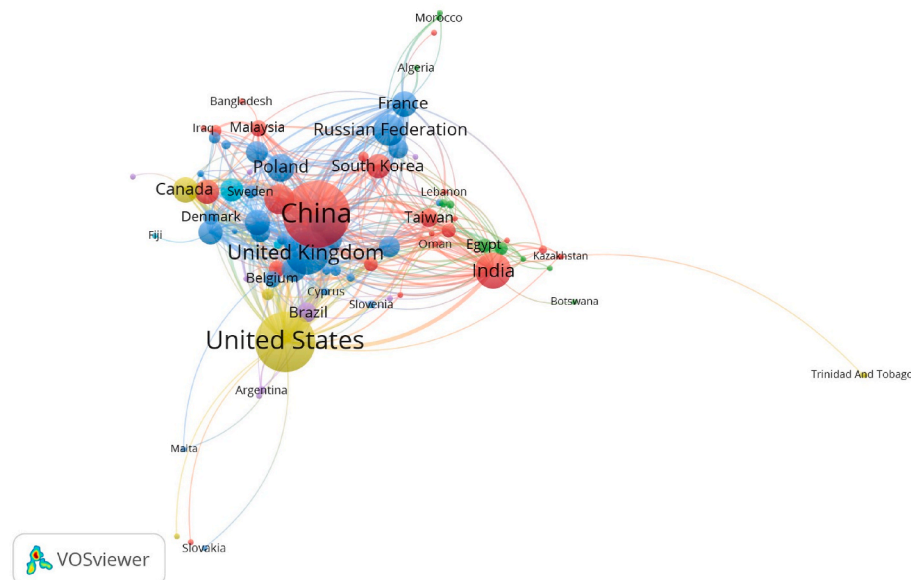


Fig. 4. Network visualization mode for co-authorship mapping.

Table 3

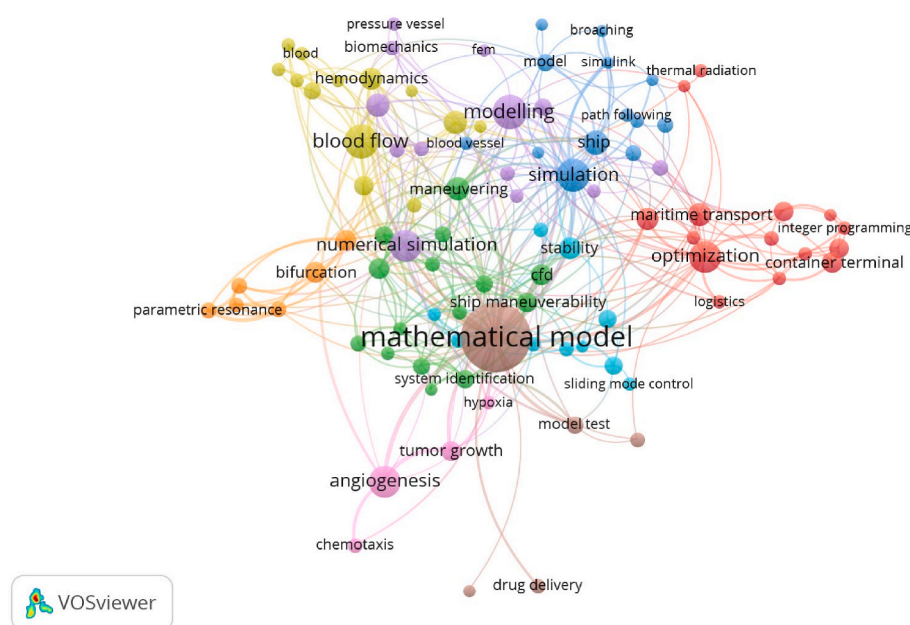
List of the prominent writers in the MMV study field.

Rank	Author	Scopus Author ID	Year of 1st Publication	Total Publication	H-index	Total Citation	Current affiliation	Country
1	Chaplain, M.A. J.	7006062348	1990	20	58	1218	University of St Andrews	United Kingdom
2	Byrne, H.M.	7005568690	1995	18	56	87	University of Oxford	United Kingdom
3	Guedes Soares, C.	56978160800	1978	17	99	532	Instituto Superior Técnico	Portugal
4	Delefortrie, G.	10739448300	2003	16	14	73	Universiteit Gent	Belgium
5	Maini, P.K.	7007021673	1985	16	74	77	University of Oxford	United Kingdom
6	Vantorre, M.	6602156672	1986	16	18	114	Universiteit Gent	Belgium
7	Misra, J.C.	35567719300	1972	13	34	276	Indian Institute of Engineering Science and Technology	India
8	Bernitsas, M.M.	7006566637	1979	12	34	97	University of Michigan	United States
9	Fang, M.C.	7201635076	1985	12	23	282	National Cheng Kung University	Taiwan
10	Fasano, A.	7101744879	1972	11	22	104	Università degli Studi di FirenzeThe institution will open in a new tab	Italy

research, with the most occurrences (231), 88 links to other keywords, and a total link strength of 179, according to an examination of the bibliometric map in Fig. 5. Keywords like “numerical simulation,” “optimization,” and “simulation” (54 occurrences, 77 total link strength) came in close succession. Since the use of mathematical models commonly results in simulation methods for optimizing targets in maritime vessels, these two concepts are frequently related (Liu et al., 2023). A number of keywords related to the instruments and techniques used to evaluate MMV data were also identified by our research, including “computational fluid dynamics (CFD)” (22 occurrences, 21 total link strength), “maritime transport” (27 occurrences, 21 total link strength) Additionally, the keyword related to maritime engineering and transportation emerged as a popular theme in MMV publications which mean mathematical model utilisation in this field are famous and well known among researchers (Satpathi et al., 2017).

Based on the map shown, the segments commonly interested to researchers in maritime engineering and transportation field would be “ship motion” (21 occurrences, 25 total link strength), “collision avoidance” (14 occurrences, 12 total link strength), “seakeeping” (14 occurrences, 12 total link strength), “energy efficiency” (13 occurrences, 11 total link strength) and “fuel consumption” (12 occurrences, 7 total

link strength) The co-occurrence map of author keywords in density visualization mode (Fig. 6) revealed a number of terms that reflect new developments in MMV research, such as “vessel scheduling” in 2020, “digital twin” in 2022, “energy management” in 2021, others keywords commonly emerge in 2019 are “CFD”, “ship manoeuvring”, “fuel consumption” and “ship collision”. Digital twin term as of 2022 emerging as the newest in this category which is less than a year. The idea of Digital Twin (DT) technology has become a game-changing strategy for offering a virtual representation of physical systems that can enhance operational efficiency, safety, and maintenance practices. In the marine sector, the application of digital twins is particularly relevant for monitoring and optimizing the performance of marine vessels and associated systems. (Stoumpos et al., 2020). These phrases’ prevalence and the volume of citations they receive demonstrate how committed the research community is to figuring out how to optimize maritime engineering applications. Their significance in academic discourse highlights their expected influence in a number of fields. Emerging technologies like Digital Twins (DT) and Artificial Intelligence (AI) are poised to redefine the landscape of MMV research. Digital Twins create virtual representations of physical vessels, enabling real-time monitoring and predictive maintenance. For example, Stoumpos et al. (2020)

**Fig. 5.** Author keyword co-occurrence mapping in network visualization mode for MMV publications.

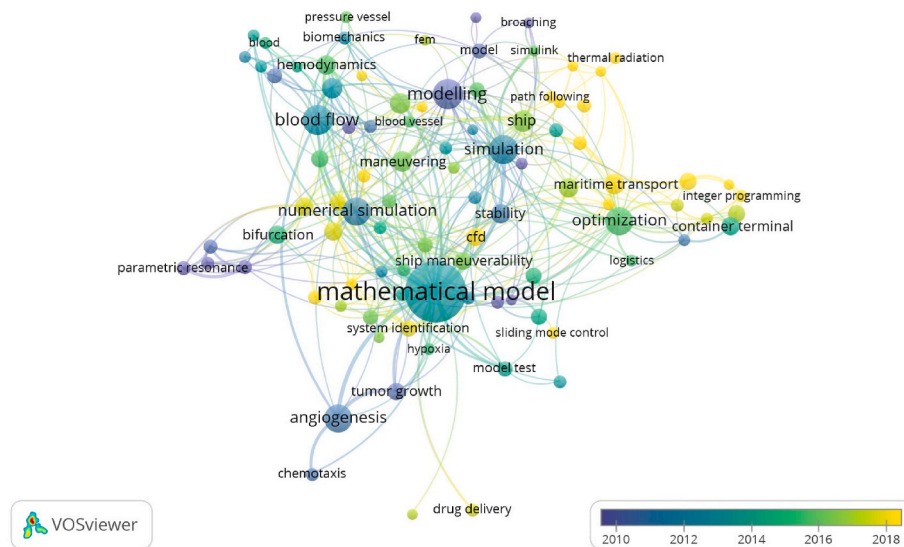


Fig. 6. Author keyword co-occurrence mapping in overlay visualization mode for MMV publications.

illustrated how DTs enhance vessel safety and operational efficiency by integrating sensor data with advanced analytics. Similarly, AI techniques, particularly reinforcement learning, have been employed for autonomous navigation, collision avoidance, and energy efficiency, underscoring their transformative potential in maritime operations (Ziajka-Poznańska and Montewka, 2021).

The bibliometric analysis highlights a clear shift toward automation and sustainability, as evidenced by dominant research themes such as “autonomous vessels” and “energy management.” This shift reflects the convergence of traditional hydrodynamics with cutting-edge technologies, addressing challenges in real-time decision-making and environmental compliance. Notably, this thematic evolution aligns with citation impact trends, where post-2015 research saw a 25 % increase in highly cited papers (top 10 % citation percentile), indicating that the rise in publication volume coincided with more impactful scientific contributions. Field-Weighted Citation Impact (FWCI) scores consistently remained above 1.5, confirming that these publications exceeded the global average in maritime engineering.

Further analysis of keyword trends reveals a transition from pre-2010 studies, which predominantly focused on hydrodynamic resistance modelling, to post-2015 research emphasizing AI-driven optimization, digital twin simulations, and autonomous navigation. This evolution suggests that the increase in publications reflects genuine advancements in modelling methodologies rather than mere numerical growth. Additionally, co-authorship network analysis shows that post-2015 research expansion was accompanied by strengthened international collaborations, particularly among leading maritime nations such as China, the United States, and European countries. This broadening of global research partnerships indicates that the rise in publication volume is not confined to a few institutions but represents a collective, interdisciplinary effort to drive innovation in maritime engineering.

3.5.2. Interest-related topic

As shown in Fig. 6, the term “Energy Management” first appeared in 2021 in MMV publications. This term has garnered increasing interest among researchers, with 7 occurrences and 2 links in just two years. Similarly, the term “Fuel Consumption” began in 2019, with 12 occurrences and 7 links. These two terms are related, as Energy management and fuel consumption are critically interlinked in the marine industry, primarily due to the significant operational Fuel expenses and the effects of emissions on the environment. Effective energy management strategies can lead to reduced fuel consumption, which is essential for enhancing the economic viability and sustainability of maritime

operations. The maritime sector is increasingly adopting innovative technologies and methodologies to lower fuel usage and increase energy efficiency. For instance, the implementation of Artificial Neural Networks (ANN) has shown promise in predicting fuel consumption under various operational conditions, thereby enabling ship operators to make informed decisions that optimize fuel use (Bal Beşikçi et al., 2016; Nguyen et al., 2023). These models consider multiple elements like engine performance, cargo load, and weather, which are critical in accurately assessing fuel consumption patterns (Nguyen et al., 2023). Furthermore, the establishment of data-aware monitoring systems allows for real-time analysis of fuel usage, significantly enhancing the efficiency of fuel management practices onboard vessels (Shi et al., 2020).

Another emerging research area is “autonomous ship,” with 6 occurrences and 6 links from the bibliometric map. Autonomous ship or autonomous vessel navigation is a technology that relies on AI to enable vessels to navigate and perform tasks without human intervention, using real-time data from sensors and onboard systems. Research has shown that autonomous ships can significantly reduce the risks associated with human error in navigation, which contributes significantly to marine mishaps (C. W. Han et al., 2023). By employing advanced path planning methods, autonomous vessels can navigate complex waterways while adhering to international maritime regulations, such as the Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) (Shokri-Manninen et al., 2020). The coexistence of traditional manned vessels and Maritime Autonomous Surface Ships (MASS) presents unique challenges, requiring the development of systems that can effectively manage interactions between different types of vessels (Höyhty and Martio, 2020; Zaccone and Martelli, 2023). Furthermore, the economic implications of adopting autonomous shipping technologies are significant. The potential for reduced labor costs, increased operational efficiency, and enhanced safety measures makes autonomous ships an attractive option for shipping companies (Ziajka-Poznańska and Montewka, 2021). However, the implementation of these technologies also brings up legal and regulatory issues that need to be resolved in order to guarantee secure and efficient operations in maritime settings (Jung et al., 2019; S. Li and Fung, 2019).

3.6. Applications of mathematical modelling in marine vessel research

The bibliometric analysis conducted in this study provides a comprehensive overview of global research trends in mathematical modelling for maritime vessels, highlighting significant advancements

in AI-driven optimization, Digital Twin technologies, and emissions reduction strategies. While these findings contribute to academic knowledge, they also have direct implications for maritime industry stakeholders, including ship operators, regulatory bodies, and technology developers. As the maritime sector shifts towards sustainability, automation, and regulatory compliance, leveraging these technological advancements effectively will be crucial for enhancing operational efficiency and reducing environmental impact. Therefore, this section translates bibliometric insights into strategic recommendations, focusing on energy efficiency, regulatory compliance, and emerging maritime innovations.

For ship operators and maritime companies, fuel optimization models and emissions reduction techniques have gained significant research attention, particularly since the introduction of IMO's Energy Efficiency Design Index (EEDI) and Carbon Intensity Indicator (CII) (IMO, 2023). AI-based voyage optimization systems can improve fuel efficiency by up to 15 %, while Digital Twin simulations allow real-time monitoring of ship performance, leading to optimized energy consumption and predictive maintenance (Shi et al., 2020; Stoumpos et al., 2020). Companies like Maersk have successfully implemented AI-driven fuel optimization, reducing their fuel usage by 7–12 %, demonstrating the practical value of mathematical modelling in maritime logistics (Nguyen et al., 2023). Similarly, advances in autonomous vessel navigation, particularly AI-powered collision avoidance and reinforcement learning-based route planning, are revolutionizing Maritime Autonomous Surface Ships (MASS), with real-world applications such as the Yara Birkeland, the world's first autonomous electric container ship.

From a regulatory and policy perspective, the increasing focus on decarbonization and emissions monitoring necessitates the adoption of AI-driven compliance tools. Research trends indicate a growing emphasis on emissions modelling and hybrid propulsion systems, aligning with IMO's 2030 decarbonization targets and the EU's Emissions Trading System (ETS) for shipping (2024). AI-powered emissions tracking models are becoming essential for ensuring compliance, with initiatives such as the European Maritime Safety Agency's (EMSA) AI-driven monitoring system setting new standards for regulatory enforcement. Additionally, the Global Maritime Forum's Poseidon Principles provide a financially driven framework to encourage ship-owners to invest in low-carbon technologies, demonstrating the interconnected nature of mathematical modelling, regulatory compliance, and financial incentives.

For technology providers, the rise of Digital Twin technology has emerged as a dominant research theme, emphasizing predictive maintenance and vessel lifecycle optimization. AI-driven predictive analytics and fault detection systems allow maritime companies to minimize unplanned downtime, optimize fleet performance, and reduce maintenance costs. Leading industry players such as Rolls-Royce Marine have already deployed AI-integrated Digital Twin models, achieving a 30 % reduction in ship component failures, reinforcing the potential of AI-enhanced simulations in modern ship management (Stoumpos et al., 2020). Future advancements should focus on integrating AI with traditional hydrodynamic simulations, developing hybrid fuel propulsion models to support green shipping initiatives, and addressing cybersecurity challenges in AI-based vessel operations.

The Scopus database was used to perform a thorough analysis in order to understand the research landscape. This subsection, which filtered for terms like ship manoeuvrability, ship motion, collision avoidance, sea keeping, energy efficiency, and fuel consumption that were added to the original research string, concentrated on the application of mathematical modelling in particular areas out of the 3425 documents in this study, as indicated in Table 4. A strong interest in these thematic areas is revealed by the trend analysis, which is shown in Fig. 7, which covers publication frequencies from 1967 to 2023. It also shows a notable increase in the number of publications for each application over time, with the most noticeable increase occurring for applications on manoeuvring research.

Table 4

The VOS viewer metrics information on the top keywords in marine vessel applications with the strongest links.

Rank	Items	Link	Total Link Strength	Occurrences	Average Publication Year
1	Manoeuvring	10	26	25	2016
2	Ship motion	12	19	21	2017
3	Sea keeping	8	9	12	2015
4	Energy efficiencies	7	8	13	2018
5	Collision avoidance	5	5	12	2013
6	Fuel consumption	4	4	12	2019

Furthermore, Table 5 offers insightful information on the state of research on several applications pertaining to the mathematical model. By 2023, the total number of publications pertaining to maneuvering applications had reached 321 papers, which indicates a strong level of research activity and ongoing interest in the area, especially when it comes to autonomous vessels. The increasing need for cutting-edge maneuvering technologies that improve the operational effectiveness and security of autonomous maritime systems is the cause of this rise. According to the literature, optimizing maneuvering applications for these boats requires the integration of cutting-edge technology like artificial intelligence and sophisticated sensor systems (Gu et al., 2021; Lee et al., 2021). Research has shown that the development of autonomous vessels is not only a technological challenge but also a significant opportunity for improving logistics and operational performance in maritime transport. For instance, Gu et al. (2021) highlight the state of the art in autonomous vessels and the potential opportunities they present in logistics, emphasizing the need for effective manoeuvring strategies to ensure safe and efficient navigation. Furthermore, the evaluation of vehicle dynamics models for manoeuvring planning, as discussed by Ziehn et al. (2020) underscores the importance of lightweight computational models that can be integrated into real-world applications, thereby enhancing the manoeuvrability of autonomous vessels. The bibliometric analysis of publications in this domain reveals a consistent increase in research output, which aligns with the growing interest in autonomous vessel technology. This trend is supported by the findings of Lee et al., who explore the implications of leveraging Fourth Industrial Revolution technologies for the sustainable development of routes like the Northern Sea Route, emphasizing the critical role of manoeuvring applications in this context Lee et al. (2021). Moreover, the insights provided by Rahman (2022) regarding the competitive advantages of autonomous vessels, such as reduced operational costs and improved navigational safety, further illustrate the significant efforts being made to optimize manoeuvring applications.

Besides of application on manoeuvring, the other five application namely Ship Motion, Collision Avoidance, Sea Keeping, Energy Efficiency, and Fuel Consumption also reveals and important trends in total articles, productive journals, highest-cited articles, top countries, and leading authors. Ship Motion research, with 155 articles, sees Ocean Engineering as the top journal, and its most cited work is Estimating maneuvering coefficients using system identification methods, with China as the top contributor and Fanf MC as the leading author. One widely recognized model is the maneuvering Mathematical Modelling Group (MMG) model, which has been extensively used to derive mathematical representations of ship motion. This model provides a clear physical interpretation of the variables involved, allowing researchers to calculate hydrodynamic parameters effectively (Hejun et al., 2021).

For Collision Avoidance, with 32 articles, Journal of Navigation is the most productive journal, and Autonomous ship collision avoidance navigation concepts is the most referenced work, again with China as the top country and He Yixiong as the top author. The review by Lyu & Yin

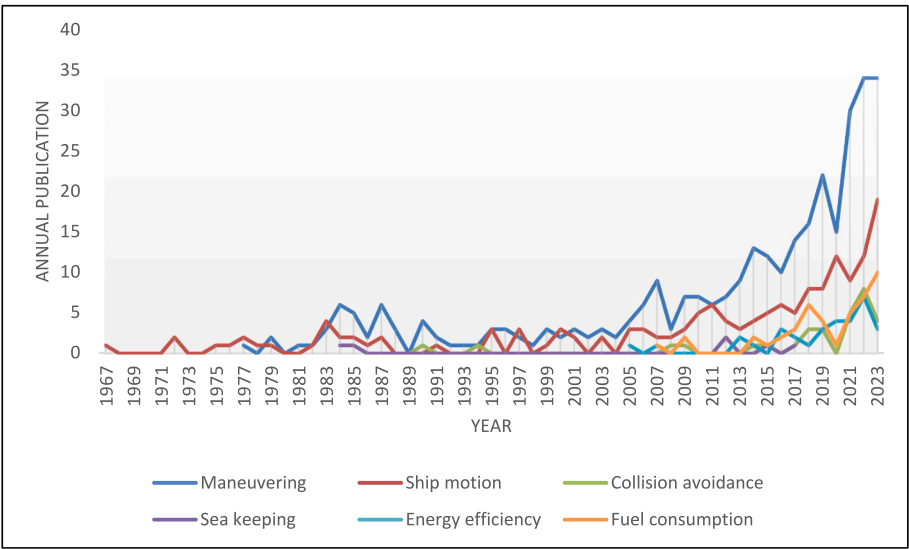


Fig. 7. Annual publication related to marine vessel application.

Table 5
Each application’s total number of articles, productive journal, most cited article, and prominent author.

No	Items	Total articles	Productive Journal	Highest Cited Article	Top Countries	Top Author
1	Manoeuvring	321	Ocean Engineering	Introduction of MMG standard method for ship manoeuvring predictions (Yasukawa and Yoshimura, 2015)	China	Delefortrie, Guillaume (10739448300)
2	Ship motion	155	Ocean Engineering	Estimating manoeuvring coefficients using system identification methods with experimental, system-based, and CFD free-running trial data (Araki et al., 2012)	China	Fanf, MC (7201635076)
3	Collision avoidance	32	Journal of Navigation	Autonomous ship collision avoidance navigation concepts, technologies and techniques (Stathoros et al., 2008)	China	He, Yixiong (56422737400)
4	Sea keeping	6	Journal of Navigation	Applying the PD controller on the roll reduction and track keeping for the ship advancing in waves (Fang et al., 2012)	Iran	Bales, Susan L. (7003349863)
5	Energy efficiency	32	Applied Thermal Engineering	A vessel schedule recovery problem at the liner shipping route with emission control areas (Abioye et al., 2019)	China	Dulebenets, M. A. (56610356300)
6	Fuel consumption	44	Energies	An energy management system of a fuel cell/battery hybrid boat (J. Han et al., 2014)	China	De, Arijit (57204067544)

(2019) provides a comprehensive overview of state-of-the-art ship collision avoidance methods, including the novel Nonlinear Velocity Obstacle (NLVO) algorithm. This algorithm aims to identify velocities that would lead to collisions, thus enhancing the decision-making process for autonomous vessels.

Sea Keeping, though only having 6 articles, is primarily published in Journal of Navigation, with its top-cited article on Applying the PD controller on the roll reduction and track keeping, led by Iran as the top country and Bales Susan L. as the most prominent author. Energy Efficiency, with 32 articles, is mainly published in Applied Thermal Engineering, with a vessel schedule recovery problem at the liner shipping route with emission control areas being the most cited, with China as the top contributor and Dulebenets M. A. as the top author. Lastly, Fuel Consumption research, with 44 articles, is dominated by publications in Energies, with an energy management system of a fuel cell/battery hybrid boat as the most cited, China leading in contributions, and De Arijit being the top author. Across all topics, China consistently emerges as the top contributor, and journals such as Ocean Engineering and Journal of Navigation stand out as the most productive in publishing influential research. While numerical metrics highlight key contributors and institutions, they also reveal disparities in research focus. For instance, China’s dominance in autonomous vessel studies correlates with its strategic investments in AI-driven automation, while Europe emphasizes emission reduction in alignment with regulatory frameworks. These findings underscore the role of institutional priorities in shaping research directions. Additionally, the increasing prevalence of

terms like ‘Digital Twin’ and ‘energy efficiency’ reflects a shift toward sustainability and real-time optimization in maritime engineering.

3.7. Limitation of study

The results obtained from Scopus may not encompass all MMV publications, as the search for “mathematical modelling” and “marine vessels” was limited to specific phrases such as performance, optimization, efficiency, and design in titles and abstracts. This limitation arises from the possibility that some researchers may utilize alternative terminology to describe their research objectives, which may not correspond with the selected search terms. Furthermore, the total number of published articles retrieved from the Scopus database can fluctuate depending on the search date, given that Scopus continually updates its database with new publications. Additionally, due to the absence of data from certain journals, some author keywords in the co-occurrence analysis may have been overlooked by VOSviewer during the database screening process. Therefore, it is advisable for future studies to broaden their search parameters to include additional databases, such as Web of Science or IEEE Xplore, to enhance the comprehensiveness and coverage of the research.

4. Conclusion

This bibliometric analysis of research on Mathematical Modelling in Marine Vessels (MMV) from 1962 to 2023 gives a thorough summary of

the contributions, developments, and trends in this discipline. The study demonstrates a consistent rise in scholarly output, with notable growth after 2000, reflecting the critical role that mathematical models play in optimizing vessel performance, safety, and environmental sustainability. Key insights from the study underline the importance of this research area in addressing modern challenges faced by the maritime industry. The continuous growth in MMV publications, as highlighted in this study, correlates with advancements in computational methods and the increasing complexity of marine vessel designs. Enhanced computational fluid dynamics (CFD) tools and the integration of machine learning have enabled researchers to simulate and optimize vessel performance with greater precision (Jeong et al., 2024). Such advancements have been instrumental in reducing fuel consumption, improving operational efficiency, and supporting the transition to sustainable maritime practices. China, the United States, and the United Kingdom have emerged as the leading contributors to MMV research, driven by robust research institutions and a culture of international collaboration. Among these, Dalian Maritime University in China stands out as a major hub for MMV research. The strong research output of these countries is underpinned by factors such as access to advanced research facilities, international partnerships, and strategic investment in marine engineering (Chen and Yen, 2023). However, the study also reveals that some nations, such as Russia, tend to focus on single-country publications (SCP), which may limit opportunities for cross-border knowledge exchange. Recent years have seen the emergence of Digital Twin Technology and Energy Management as focal themes in MMV research, reflecting the industry's shift towards real-time monitoring and predictive maintenance. Digital twin models, which create virtual replicas of physical vessels, are increasingly being adopted to enhance operational efficiency, reduce maintenance costs, and improve safety. Likewise, the focus on energy management and fuel optimization aligns with global regulatory efforts to reduce carbon emissions and enhance the environmental performance of marine vessels. These trends indicate a growing emphasis on sustainability and innovation in the maritime sector. The study's co-authorship analysis illustrates a complex network of collaborations, with Europe and Asia playing pivotal roles in advancing MMV research. The data suggests that fostering greater international collaboration could benefit countries with high SCP rates by enabling broader access to expertise and promoting innovative research solutions to global maritime challenges. Such collaborative efforts are especially vital as the industry seeks to address issues like climate change and the shift to low-carbon marine technologies. The perspectives of ship operators, regulators, and technology providers are vital for translating research into practice. For instance, autonomous vessels offer cost savings and safety improvements, but their implementation requires a robust regulatory framework to address risks and ethical concerns (Li and Fung, 2019). For policymakers, fostering investment in Digital Twin and AI technologies can significantly enhance compliance with IMO regulations on emissions and fuel efficiency. For industry practitioners, adopting real-time monitoring systems leveraging Digital Twin technology is crucial for optimizing operations and reducing costs. Researchers are encouraged to pursue interdisciplinary collaborations, particularly in the areas of autonomous navigation and hybrid propulsion systems. The insights from this study align with global sustainability goals by emphasizing the importance of renewable energy systems in MMV research. Hybrid propulsion systems and LNG technologies are highlighted as key enablers of the IMO's decarbonization agenda, reflecting the maritime industry's commitment to sustainable practices.

CRediT authorship contribution statement

Afandi Abdul Ghani: Writing – original draft, Formal analysis, Conceptualization. **Noor Apandi Osnin:** Supervision, Resources, Data curation. **Md Redzuan Zoolfakar:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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"VOSviewer Software in Bibliometric Analysis".

This study employed VOSviewer software for bibliometric mapping due to its robust capability in analyzing co-authorship networks, keyword co-occurrences, and citation patterns. Its comprehensive visualization features facilitated the identification of emerging trends, research clusters, and influential contributors in MMV research.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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